Intel Exchange

Cyware Intel Exchange - Google SecOps SIEM Integration

User Guide
Version: 1.0.0

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Overview

Cyware Intel Exchange Platform

Cyware Intel Exchange is an AI-powered threat intelligence platform that provides comprehensive security orchestration, automation, and response capabilities. The platform enables organizations to aggregate, analyze, and operationalize threat intelligence while automating security operations. Cyware Intel Exchange's vendor-neutral, low-code solution allows security teams to optimize operations, automate cross-functional workflows, and accelerate threat response across the entire security infrastructure.

Google SecOps

Google SecOps is a cybersecurity telemetry platform for threat hunting, and threat intelligence and is part of the Google Cloud Platform. Google SecOps stores log events it receives in two formats: either as the original raw log or structured Unified Data Model (UDM) log. There are two critical elements to consider for parsing, Unified Data Model (UDM) which defines the schema for parsing, and Configuration Based Normalizers (CBN) which describes how log data is transformed to the UDM schema.

Cyware Intel Exchange Integration for Google SecOps SIEM Integration

This integration enables the seamless ingestion, parsing, and visualization of Cyware Intel Exchange data within Google SecOps SIEM, offering comprehensive insights through native Google SecOps dashboards. It supports real-time delivery of Indicators of Compromise (IOCs) from Cyware Intel Exchange platform, automatically enriching security events with contextual threat data. The integration provides security teams with immediate visibility into threats relevant to their environment, enabling faster detection and response to potential security incidents.

Release Notes

V1.0.0

- Provided the ingestion script which can deploy as a GCP cloud function to collect data from the Cyware Intel Exchange Platform and ingest in the Google SecOps.
- Provided the parser that processes data ingested from the Cyware Intel Exchange platform and converts it into the Google SecOps UDM data model.
- Provided below dashboards for visualization:
    - Cyware Intel Exchange - Cyware IOCs Overview
    - Cyware Intel Exchange - Correlation Overview
- Provided below correlation rules:
    - Brute Force Attempt
    - Malicious Domain Detection
    - Malicious Email Sender Detection
    - Malicious File Hash MD5 Detection
    - Malicious File Hash SHA1 Detection
    - Malicious File Hash SHA256 Detection
    - Malicious IP Detection
    - Malicious URL Access Detection
- For Cyware Intel Exchange integration, we have introduced support for log parsing under the new **CTIX** label.

App Installation & Configuration

Pre-Requisites

- Google SecOps
- Cyware Intel Exchange credentials (Access ID and Secret Key)
- GCP Project with the below required permissions:
    - GCP user and project service account should have below permissions. Follow these steps to assign these permissions to the service account used for accessing other GCP services within your project.
        - Storage Admin
        - Chronicle API Editor
        - Cloud Run Admin
        - Project IAM Admin
        - Secret Manager Admin
        - Service Account User
        - Cloud Functions Developer
        - Cloud Scheduler Admin
        - Secret Manager Secret Accessor
        - Cloud Scheduler Job Runner
- GCP Services
    - Before using the below services, ensure the Google APIs are enabled in your GCP project
        - Cloud Run Function (4-core CPU or higher is recommended)
        - GCS bucket
        - Secret Manager
        - Cloud Scheduler
- Access to Google SecOps Rules & Detection section.
- Access to Google SecOps Dashboard section.

Creating zip of the cloud function

- Create a zip file with the contents of the following files:
    1. Download the common directory from Git repository.
    2. Download the contents of the ingestion script from Git repository.
    3. Create a zip in the following structure.

❖ Example:

```
├── README.md
├── common
│   ├── __init__.py
│   ├── auth.py
│   ├── auth_test.py
│   ├── env_constants.py
│   ├── ingest.py
│   ├── ingest_test.py
│   ├── ingest_v1.py
│   ├── ingest_v1_test.py
│   ├── status.py
│   ├── utils.py
│   └── utils_test.py
├── constant.py
├── exception_handler.py
├── exception_handler_test.py
├── main.py
├── main_test.py
├── requirements.txt
├── utility.py
├── utility_test.py
├── cyware_client.py
└── cyware_client_test.py
```

Automatic deployment of the required resources

This section explains the use of the provided bash script to automate the deployment of the Cyware Intel Exchange ingestion function and related resources within your GCP project.

Prerequisites For Script Execution :

- Access to a GCP project with sufficient permissions to enable APIs, create service accounts, IAM bindings, GCS buckets, Cloud Functions, Cloud Scheduler jobs, and Secret Manager secrets.
- The Cloud Function source code packaged as a ZIP file in this format, available on your local machine or accessible within the Cloud Shell environment.

Execution Steps :

1. **Open Google Cloud Shell :**
   - Navigate to the Google Cloud Console (console.cloud.google.com).
   - Click the "**Activate Cloud Shell**" in the upper right corner of the console. A terminal window will open within your browser.
2. **Create the Script File :**
   - Choose a name for the script, for example, `cyware_deploy.sh`
   - Use a text editor like `nano` to create the file in your Cloud Shell home directory: `nano cyware_deploy.sh`
3. **Paste the Script Content :**
   - Copy the entire content of the bash script provided from github repo.
   - Paste the script content into the `nano` editor. In Cloud Shell, you can typically right-click and select "Paste" or use `Ctrl+Shift+V.`
4. **Save and Exit :**
   - Press `Ctrl+X` to exit `nano.`
   - When prompted to save, press `Y`.

- Press `Enter` to confirm the filename (`cyware_deploy.sh`)

5. **Upload Your Cloud Function ZIP (if not already in Cloud Shell) :**
    - If your function ZIP file is on your local machine, you need to upload it to Cloud Shell.
    - Click the three-dot menu in the Cloud Shell terminal window.
    - Select "Upload" and choose your ZIP file. It will be uploaded to your Cloud Shell home directory. Note the filename.
6. **Make the Script Executable :**
    - Grant execute permissions to the script file: `chmod +x cyware_deploy.sh`
7. **Run the Deployment Script :**
    - Execute the script: `./cyware_deploy.sh`
8. **Follow the Prompts :**
    - The script will prompt you to enter various configuration details, such as:
        - GCP Project ID
        - GCP Region
        - Local path to your Cloud Function ZIP file (e.g., ./your-function.zip if it's in the home directory)
        - Chronicle Customer ID
        - Chronicle Project Number
        - Chronicle Region
        - Local path to Chronicle Service Account (Required if you are opting for a custom service account authentication option.)
        - Cyware Base URL (e.g., https://your-ctix-instance.com/ctixapi/)
        - Cyware Tenant Name
        - Cyware Access ID Value (this will be stored securely in Secret Manager)

    - Cyware Secret Key Value (this will be stored securely in Secret Manager)
    - Enable Enrichment of Indicators (true or false)
    - Cyware Indicators Lookback Days
    - Cyware Saved Result Set Name
    - CRON Schedule (Refer this link for format)
- Provide the requested information at each prompt. Required fields must be filled. Optional fields can be left blank to use defaults where applicable. These values can be obtained by following these steps.

9. Monitor the Output :

- The script will display progress messages, indicating which steps are being performed (e.g., enabling APIs, creating resources, deploying the function).
- Error messages will be shown in red. If an error occurs, the script is designed to stop. Review the error message to troubleshoot.

10. Post-Deployment :

- Once the script completes successfully, verify the resources in the GCP Console:
    - Check Cloud Functions to see your deployed function.
    - Check Cloud Scheduler to see the scheduled job.
    - Check GCS for the bucket and upload ZIP.
    - Check Secret Manager for the Cyware Access ID and Secret Key secrets.
    - Review Cloud Logging for any function execution logs or errors.

Manual deployment of the required resources

Enabling required APIs

Ensure the following Google APIs are enabled in your GCP project (via **APIs & Services** → **Library**).

Service	**APIs needs to enable**
Google SecOps	• chronicle.googleapis.com
Cloud Functions	• cloudfunctions.googleapis.com • run.googleapis.com • cloudbuild.googleapis.com • artifactregistry.googleapis.com • logging.googleapis.com
Cloud Scheduler	• cloudscheduler.googleapis.com • pubsub.googleapis.com
Cloud Storage (Bucket)	• storage-component.googleapis.com
Secret Manager	• secretmanager.googleapis.com
Cloud Resource Manager	• cloudresourcemanager.googleapis.com

Using Secrets

- Environment variables marked as secret must be configured as secrets on Google Secret Manager. [REF]
- Once the secrets are created on Secret Manager, use the secret's resource ID as the

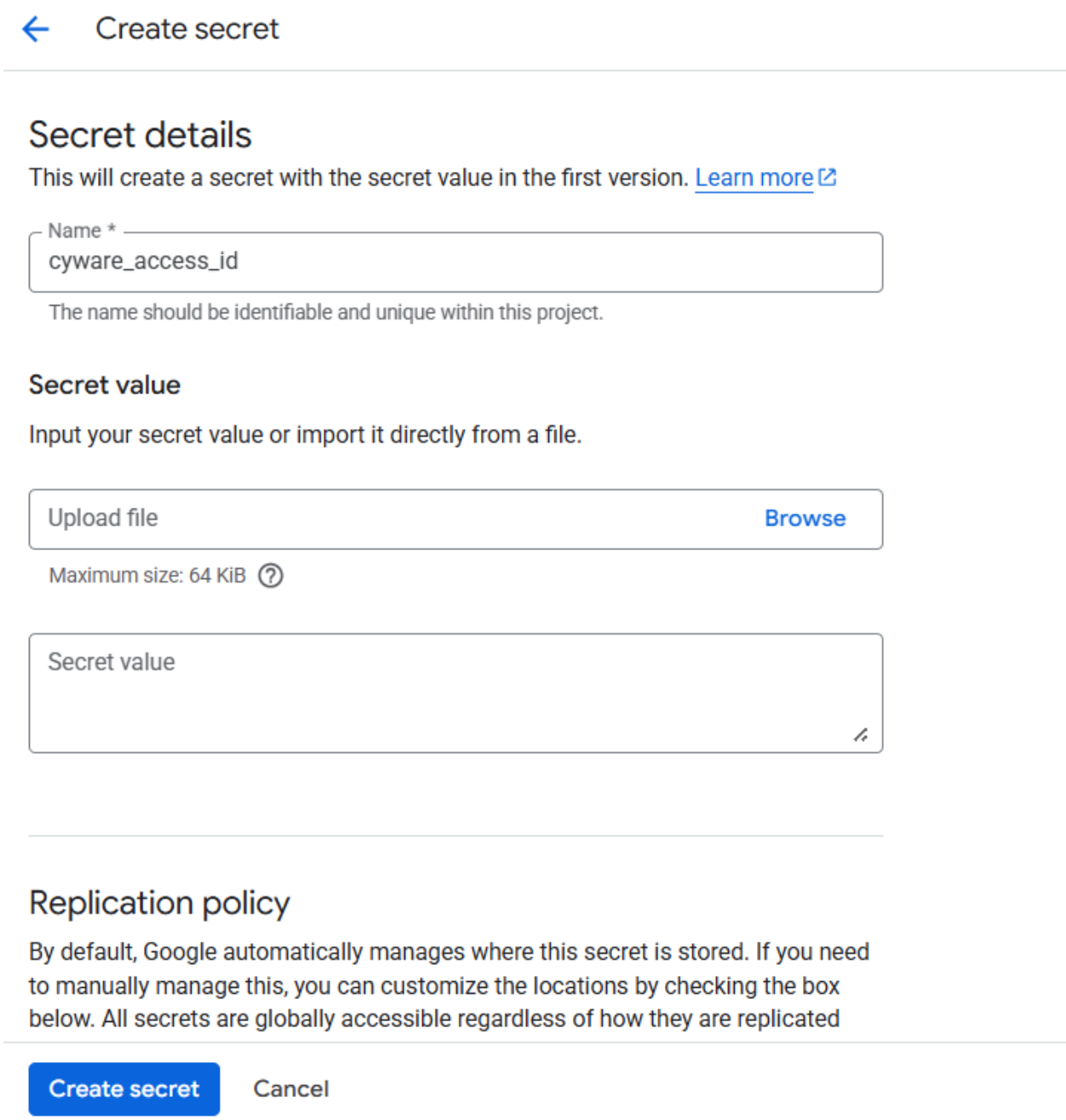
value for environment variables.

For example:
CHRONICLE_SERVICE_ACCOUNT:
projects/{project_id}/secrets/{secret_id}/versions/{version_id}

Add the Secret in Secret Manager

Follow the steps below to create separate secrets for the Cyware Access ID, Cyware Secret Key, and the Chronicle Service Account (when using a custom service account):

1. Log in to the "https://console.cloud.google.com/" using valid credentials.
2. Navigate to '**Secret Manager**'.
3. Click on '**Create Secret**'.
4. Provide the name for the secret in the `Name` field.
5. Provide the value for the secret in the `Secret Value` field.
6. Keep the other configurations as default, Click on the '**Create Secret**' button.



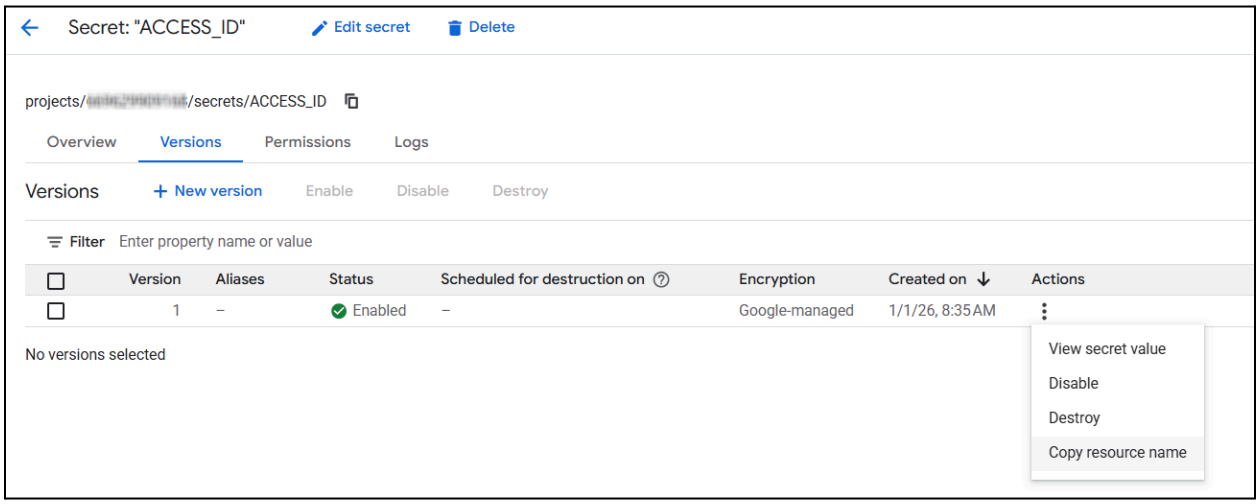
Refer to this page for more information about how to create secrets.

Once the secrets are created on Secret Manager, use the secret's resource name as the value for environment variables. For example

CYWARE_ACCESS_ID: projects/{project_id}/secrets/{secret_id}/versions/{version_id}

Steps to get Secret's Resource Name:

1. Go to the **Secret Manager** inside the GCP project.
2. Click on Secret name.
3. Click on ⋮ of a specific version.
4. Click on **Copy resource name**.

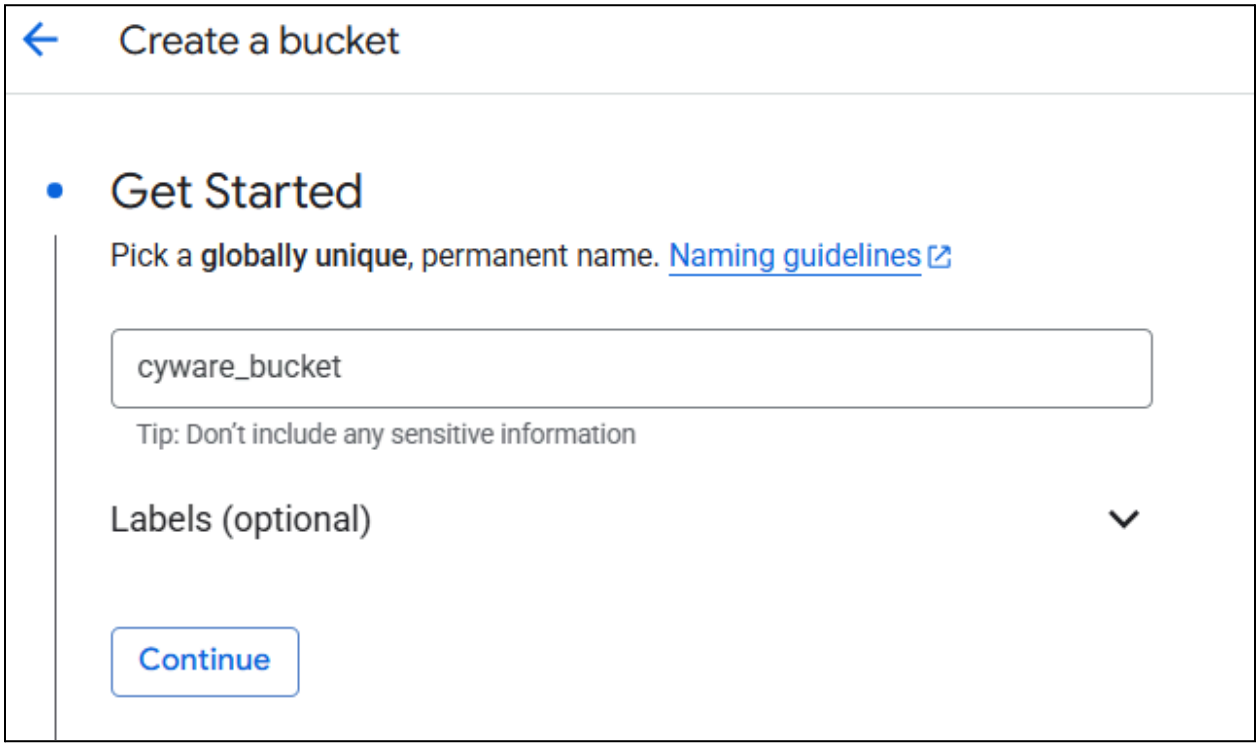


Note : Ensure that the resource Name format matches the following structure: projects/{project_id}/secrets/{secret_id}/versions/{version_id}.

Create a GCP Bucket

1. Log in to the "https://console.cloud.google.com/" using valid credentials.
2. Navigate to Buckets in GCP.
3. Click on the Create button.
4. Enter the name of the bucket.
5. Users can select the region and modify the optional parameters if required and then click on the Create button.

Copy the bucket name and provide it in the **GCP_BUCKET_NAME** environment variable.



Authentication & Permissions

The Cloud function supports authentication to Chronicle API using following two modes of authentication:

Option 1 - Default Cloud Function Service Account
If CHRONICLE_SERVICE_ACCOUNT environment variable is not set, then the function will automatically use the service account used during the Cloud Function deployment. Ensure this service account has the Chronicle API Editor role assigned.

Note: Make sure to deploy the cloud function in the same GCP project where the Google SecOps instance is configured in order to access the default service account. Otherwise consider the option 2 custom service account where you have the flexibility to deploy the cloud function to another GCP project.

Option 2 - Custom Service Account
Create a service account with Chronicle API Editor role.
Store the Service Account credentials securely in Secret Manager following these steps.
Set the CHRONICLE_SERVICE_ACCOUNT environment variable in the Cloud Function.
The function will use this service account for authenticating Chronicle API requests.

Note: If both are available, the `CHRONICLE_SERVICE_ACCOUNT` value takes priority over the default service account.

Cloud Function Deployment

Command based deployment

1. Navigate to the bucket and open the bucket created for the Cyware in these steps. Upload the created cloud function zip file in the bucket.
2. Click Activate Cloud Shell at the top right corner of the Google Cloud console.
3. Modify the below command based on your value and run in the terminal.

Command Format :

```
gcloud functions deploy CLOUD_FUNCTION_NAME --set-env-vars
"ENV_NAME1=ENV_VALUE1,ENV_NAME2=ENV_VALUE2,ENV_NAME3=ENV_VALUE3" --gen2 --runtime=python312 --region=REGION
--source=SOURCE_OF_FUNCTION  --entry-point=main
--service-account=SERVICE_ACCOUNT_EMAIL --trigger-http
--no-allow-unauthenticated --memory=8GiB --timeout=3600s
```

- **CLOUD_FUNCTION_NAME**: Unique name of the cloud function.
- **REGION**: A region for your cloud function. (Ex : us-central1, us-west1, etc.)
- **SOURCE_OF_FUNCTION**: gsutil URI of the cloud function zip in cloud storage. (e.g: gs://cyware_bucket/cyware.zip) where cyware_bucket is the name of the created bucket and cyware.zip is the cloud function zip file.
- **SERVICE_ACCOUNT_EMAIL**: Email of the created service account of the project. Make sure the selected Service account must have a required permission. Update Service Account Permission following these steps.
- **ENV_NAME1**=**ENV_VALUE1**: Name and value of the environment variable to be created. Environment variables

Note:

1. When deploying a Cloud Function, ensure that the **--timeout** parameter in the deployment command matches the frequency specified in the Cloud Scheduler **--schedule** parameter. Aligning these values prevents overlapping executions, which could lead to data duplication.
    - **For example**, if you set --timeout=3600s when deploying the Cloud Function, configure the Cloud Scheduler with: --schedule="*/60 * * * *".

- This ensures that each scheduled run starts only after the previous execution has completed.

Example Command to Deploy Cloud Function,

```
gcloud functions deploy cyware-ctix-secops-function --set-env-vars
"CHRONICLE_CUSTOMER_ID=ed19f037-2354-43df-bfbf-350362b45844,CHRONICLE
_SERVICE_ACCOUNT=projects/1234567890/secrets/chronicle_service_account/versi
ons/1,CHRONICLE_PROJECT_NUMBER=2134567,CHRONICLE_REGION=us,GCP_BUC
KET_NAME=cyware_ctix_bucket,CYWARE_ACCESS_ID=projects/1234567890/secrets
/cyware-access-id/versions/1,CYWARE_SECRET_KEY=projects/1234567890/secrets/
cyware-secret-key/versions/1,CYWARE_BASE_URL=https://ctix.example.com,CYWA
RE_ENRICHMENT_ENABLED=true,CYWARE_INDICATOR_LOOKBACK_DAYS=30,CYWA
RE_SAVED_RESULT_SET_NAME=my-saved-result-set" --gen2 --runtime=python312
--region=us-central1 --source=gs://cyware_ctix_bucket/cyware_ctix.zip
--entry-point=main
--service-account=1234567890-compute@developer.gserviceaccount.com
--trigger-http --no-allow-unauthenticated --memory=8GiB --timeout=3600s
```

Configure Scheduler

Command based deployment

1. Click Activate Cloud Shell at the top right corner of the Google Cloud console.
2. Modify the below command based on your value and run in the terminal.

Command Format :

```
gcloud scheduler jobs create http SCHEDULER_NAME --schedule="CRON_TIME"
--uri="CLOUD_FUNCTION_URL" --attempt-deadline=30m
--oidc-service-account-email=SERVICE_ACCOUNT_EMAIL --location=LOCATION
--time-zone=TIME_ZONE
```

- **SCHEDULER_NAME**: Unique name of the cloud scheduler.
- **CRON_TIME**: Cron time format for the scheduler to run in every interval. (eg. */60 * * * *)
- **CLOUD_FUNCTION_URL**: URL of the created cloud function. Navigate to create cloud function details.
- **SERVICE_ACCOUNT_EMAIL**: Email of the created service account of the project. Make sure the selected Service account must have a required Permission. Update Service Account Permission following these steps.
- **LOCATION**: A region for your connector. (Ex : us-central1, us-west1, etc)
- **TIME_ZONE**: The time zone of your region. (Ex : UTC)

Note:

1. When deploying a Cloud Function, ensure that the **--timeout** parameter in the deployment command matches the frequency specified in the Cloud Scheduler **--schedule** parameter. Aligning these values prevents overlapping executions, which could lead to data duplication.
    - **For example**, if you set --timeout=3600s when deploying the Cloud Function, configure the Cloud Scheduler with: --schedule="*/60 * * * *".
    - This ensures that each scheduled run starts only after the previous execution has completed.

Example Command to Deploy Cloud Scheduler,

```
gcloud scheduler jobs create http cyware-ctix-secops-scheduler --schedule="0 * *
* *" --uri="https://us-central1-test.cloudfunctions.net/cyware-ctix-secops-function"
--attempt-deadline=30m
--oidc-service-account-email=1234567890-compute@developer.gserviceaccount.c
om --location=us-central1 --time-zone=UTC
```

Update Service Account Permission

1. Open **GCP Console**, Then go to **IAM**.
2. In View By **Main Tab** > Click **GRANT ACCESS**.
3. Add Service Account name in **New Principals**. (Example : service_account_name.gserviceaccount.com)
4. In **Assign Role**, assign below roles to service accounts.
    a. Cloud Scheduler Job Runner
    b. Secret Manager Secret Accessor
    c. Storage Admin
    d. Chronicle API Editor
    e. Cloud Run Invoker
    f. Role Viewer
5. Click **Save**.

Environment Variables to be configured in Cloud Function

During the integration setup, users are required to provide environment specific values. These values can be obtained by following these steps.

Environment variable	Description	Default value	Required	Secret Manager
CHRONICLE_CUSTOMER_ID	Google SecOps customer id. Navigate to settings in the Google SecOps console for the customer id.	-	Yes	No
CHRONICLE_SERVICE_ACCOUNT	Copied resource name value of service account secret from the secret manager. Required if you are opting for a custom service account authentication option.	-	No	Yes
CHRONICLE_REGION	A region where the Google SecOps instance is located.	us	No	No
CHRONICLE_PROJECT_NUMBER	Specifies the GCP project identifier associated with your Google SecOps environment.	-	Yes	No
GCP_BUCKET_NAME	Name of the created GCP bucket.	-	Yes	No
CYWARE_ACCESS_ID	Copied resource name value of Access ID of Cyware Intel Exchange from secret manager. Generate Access ID from the Cyware Intel Exchange platform's Administration section. Refer link.	-	Yes	Yes

CYWARE_SECRET_KEY	Copied resource name value of Secret Key of Cyware Intel Exchange from secret manager. Generate Secret key from the Cyware Intel Exchange platform's Administration section. Refer link.	-	Yes	Yes
CYWARE_BASE_URL	Base URL of your Cyware Intel Exchange instance (e.g., https://your-ctix-instance.com/ctixapi/). Generate Base URL from the Cyware Intel Exchange platform's Administration section. Refer link.	-	Yes	No
CYWARE_TENANT_NAME	Tenant name for multi-tenant Cyware Intel Exchange deployments.	-	No	No
CYWARE_ENRICHMENT_ENABLED	Flag to enable enrichment of indicators. Accepted values [true, false]	false	No	No
CYWARE_INDICATOR_LOOKBACK_DAYS	The historical time for indicator data fetching in days.	7	No	No
CYWARE_SAVED_RESULT_SET_NAME	Filter indicators by specific label name in Cyware Intel Exchange.	-	Yes	No

Note: When using the manual deployment flow, avoid comma-separated values for CYWARE_SAVED_RESULT_SET_NAME. Use the provided deployment script if you need to supply comma-separated saved result set names.

View Entities in Google SecOps

---

1. Log in to Google SecOps:
    a. Open a web browser and navigate to the Google SecOps instance URL. For example: https://test.backstory.chronicle.security/
    b. Replace test with your actual Google SecOps instance name.
2. Access SIEM Search:
    a. From the top left corner of the Google SecOps console, select the "Investigation" option.
    b. Within the Investigation section, choose "SIEM Search".
3. Filter Entities by Log Type:
    a. In the SIEM Search interface, locate the "UDM Search" section.
    b. Apply a filter for the field "log_types". Set the filter value to `graph.metadata.event_metadata.base_labels.log_types = "CTIX"`
4. View Cyware Intel Exchange Entities:
    a. The SIEM Search results will display Cyware Intel Exchange entities within the "Results" section.

UDM Mapping Sheet

---

1. The Mapping sheet will indicate the UDM entity type and which raw log field will be mapped with the corresponding UDM field parsed by parser.
2. UDM Mappings: link

Redeploying an Upgraded Build

Run the same Automatic or Manual steps with the new Cloud Function ZIP. Use the same function name so GCP updates the code automatically.

Deployment Steps:

Automatic: Rerun the bash script to provide the **updated ZIP**, and reuse the earlier configuration values (Project ID, API key, etc.).

Manual: Rerun the gcloud functions deploy command, point the source flag to the **latest ZIP**, and keep the **function name** unchanged.

Important Notes:

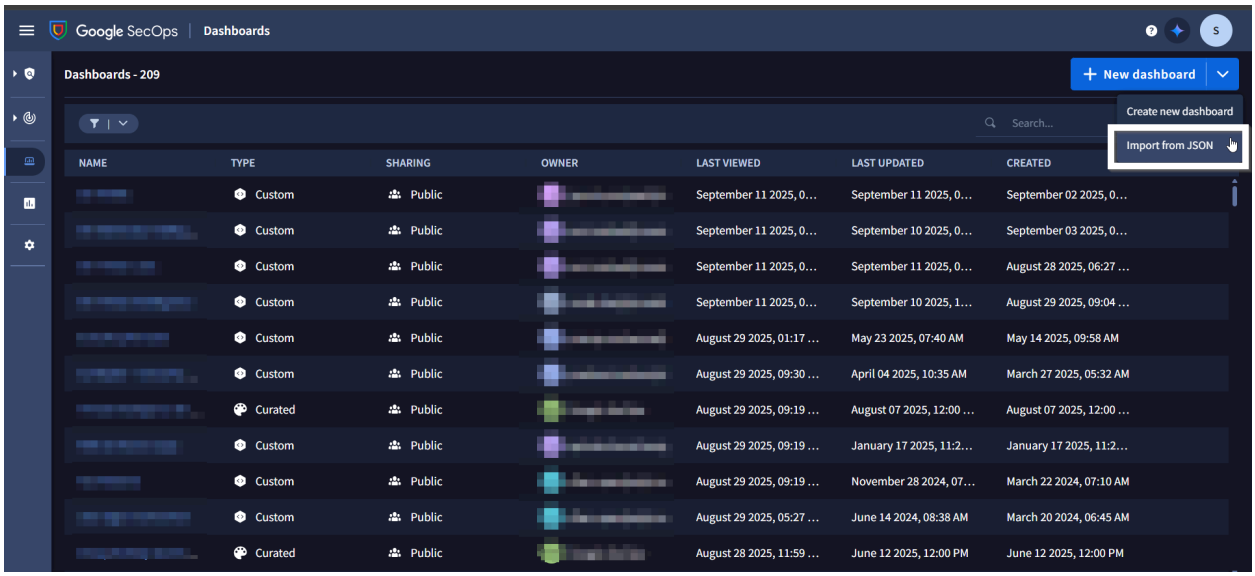
- Same function name means the old build is overwritten, no extra cleanup required.
- Cloud Scheduler, Secret Manager, and service-account settings stay valid.
- After redeploying, check Cloud Logging to confirm everything runs without errors.

Dashboard Configuration

Import Cyware Intel Exchange Dashboard into Google SecOps SIEM

Complete the following steps to import a dashboard:

1. Log in to your Google SecOps instance.
2. In the navigation bar, click **Dashboards**.
3. Click **New dashboard** and then select **Import from JSON**.



   The **Select file** dialog appears.

4. Select a file and click **Confirm**.The **Import dashboard** confirmation dialog appears.
5. Click **Import** to continue importing the dashboard to a personal or shared dashboard.

Dashboards

After collecting the Cyware data into the Google SecOps, the user can visualize the data on the Dashboard. This dashboard has the following features mentioned below.

Cyware Intel Exchange - Correlation Overview

The Correlation dashboard is designed to identify relationships of Cyware Intel Exchange's IOCs and events from other sources. The correlation dashboard is limited to support ip address, domain and file hash (MD5, SHA-1, SHA-256) only due to Google SecOps limitation.

1. Filters description

Global Time Filter

This filter updates the panels based on the time range selected in it. **Default:** Last 7 days.

2. Panels description

Unique IOC Matches

Displays number of unique Cyware IOCs that were matched in the environment within the specified global time filter.

Visualization type: Number

Unique IOC Matches (Last 30 Days)

Displays number of unique Cyware IOCs that were matched in the environment within the last 30 Days.

Visualization type: Number

Unique IOC Matches (Last 7 Days)

Displays number of unique Cyware IOCs that were matched in the environment within the last 7 Days.

Visualization type: Number

Unique IOC Matches (Last 24 Hours)

Displays number of unique Cyware IOCs that were matched in the environment within the last 24 Hours.

Visualization type: Number

High Severity IOC Matches

Display number of IOC matches which have High severity or confidence score is greater than 69 within the specified global time filter.

Visualization type: Number

Medium Severity IOC Matches

Displays number of IOC matches which have Medium severity or or confidence score is between 30 to 69 within the specified global time filter.

Visualization type: Number

IOC Matches Over Time By Severity

Displays number of IOC matches that have matched over time within the specified global time filter.

Visualization type: Line Chart

Top 10 IOC Values

Lists the top 10 IOCs that repeatedly appear in IOC matches within the specified global time filter. This helps SOC teams identify persistent attackers. It has a drilldown on IOC value view to show events that were matched to Cyware IOCs.

Visualization type: Table

IOCs Matched

Provides a detailed list of all IOCs that have matched in the environment within the specified global time filter. It has a drilldown on IOC value view to show events that were matched to Cyware IOCs.

Visualization type: table

IOCs Geolocation Overview

Shows the geographic source of the most frequent IOC matches within the specified global time filter.

Visualization type: Map

Number of IOC Matches by Risk Score

Shows IOC matches segmented by risk score ranges (None = 0, Low = 1-29, Medium = 30-69, High = 70-100) within the selected global time filter.

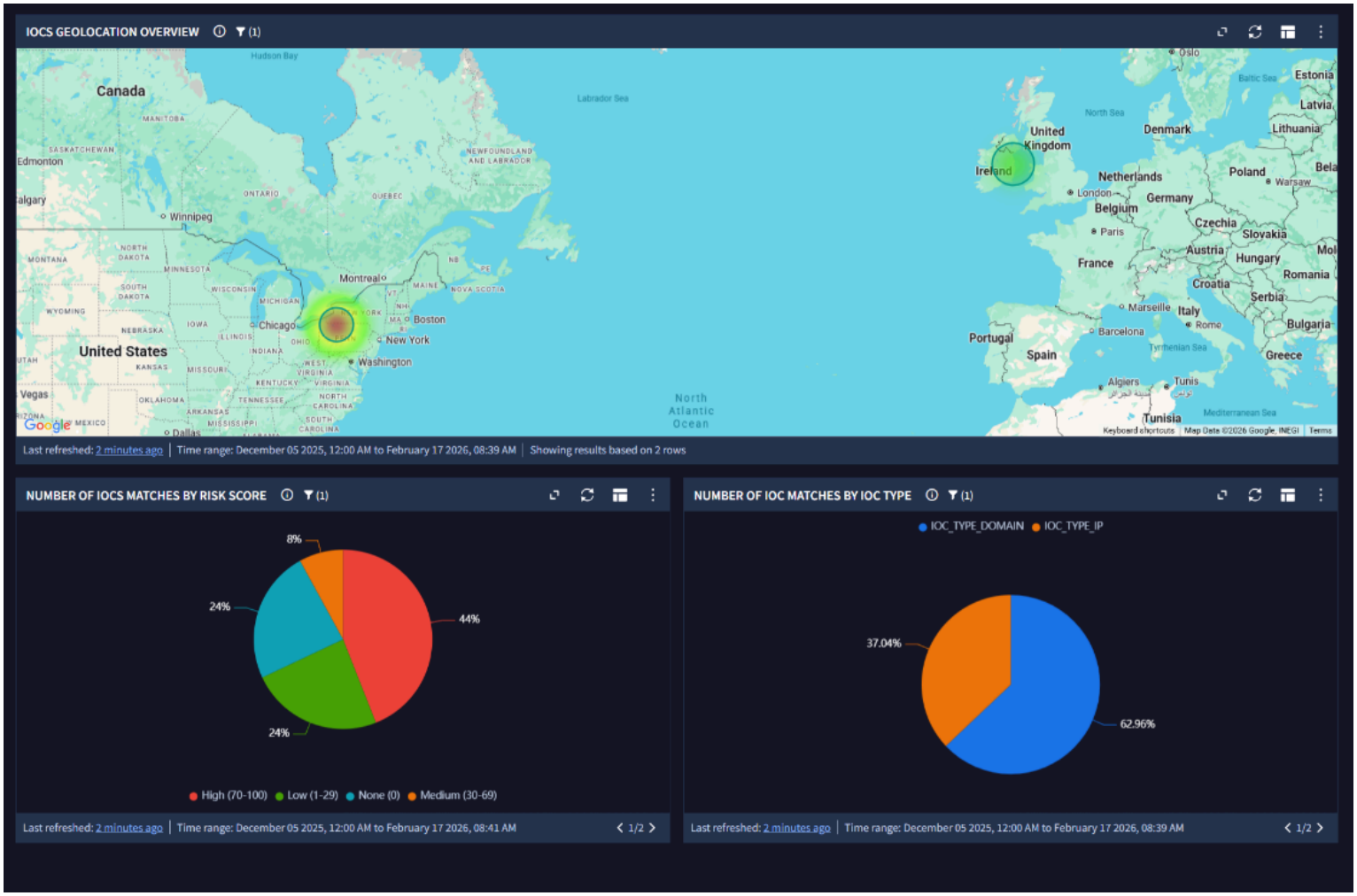
Visualization type: Pie chart

Number of IOC Matches by IOC Type

Displays the number of IOC Matches for each IOC type within the specified global time filter.

Visualization type: Pie chart

Cyware Intel Exchange - Correlation Overview - 1



Cyware Intel Exchange - Correlation Overview - 2

Cyware Intel Exchange - IOCs Overview

The dashboard is designed to visualize the IOCs ingested in SecOps from Cyware Intel Exchange.

1. Filters description

IOC Active time

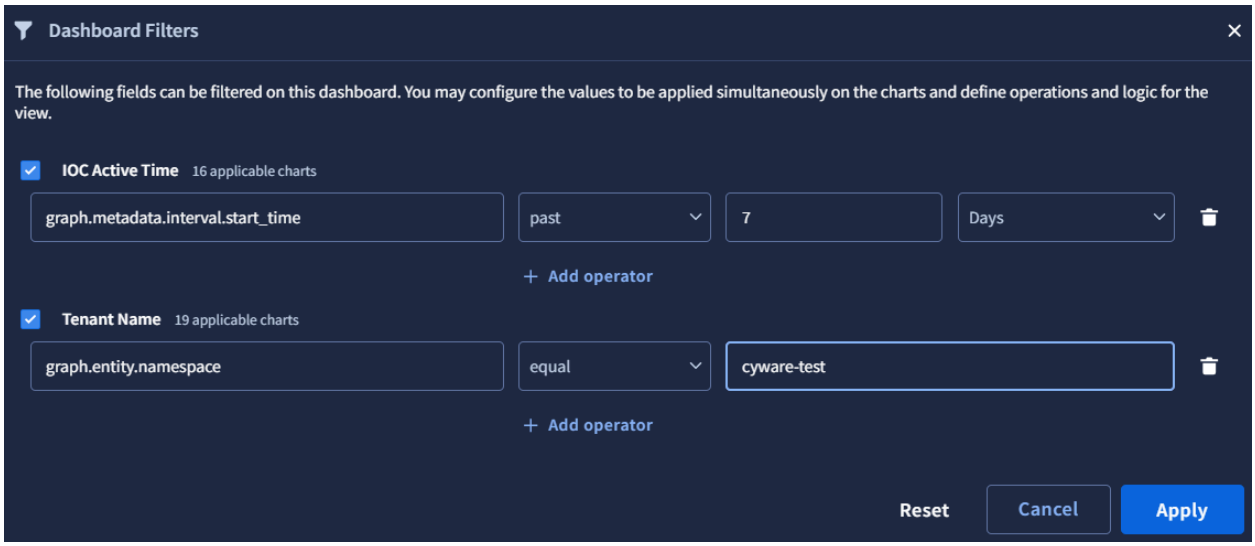
This filter updates the panels based on the time range selected in it. **Default:** Last 7 days.

Tenant Name

This filter updates the panels based on the tenant name selected. **Default:** not null

Steps to apply specific Tenant Name on Filter:

1. Click on **Tenant Name** filter
2. Set operator to **equal**.
3. Type Tenant Name in Text box.
4. Click on **Apply**.



2. Panels Description

Total Unique Cyware IOCs

Displays total number of IOCs ingested in Secops from Cyware within the specified IOC Active time filter.

Visualization type: Number

Total Unique Cyware IOCs (Last 30 Days)

Displays total number of IOCs ingested in Secops from Cyware within the last 30 Days.

Visualization type: Number

Total Unique Cyware IOCs (Last 7 Days)

Displays total number of IOCs ingested in Secops from Cyware within the last 7 Days.

Visualization type: Number

Total Unique Cyware IOCs (Last 24 Hours)

Displays total number of unique IOCs ingested in Secops from Cyware within the last 24 Hours.

Visualization type: Number

High Severity Cyware IOCs

Displays number of IOCs which have High severity or confidence score is greater than 69 within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Medium Severity Cyware IOCs

Displays number of IOCs which have Medium severity or confidence score is between 30 to 69 within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Pending Manual Review Cyware IOCs

Displays number of IOCs in Cyware that are marked for manual review. This panel helps threat intel and SOC teams focus on unreviewed indicators that may require manual review. The panel has a drilldown for IOCs.

Visualization type: Number

Reviewed Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that are reviewed within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Actioned Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that have action taken within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Revoked Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that are revoked within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Whitelisted Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that are whitelisted within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

False Positive Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that are false positive within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Deprecated Cyware IOCs

Displays number of IOCs in Cyware Intel Exchange that are deprecated within the specified IOC Active time filter. The panel has a drilldown for IOCs.

Visualization type: Number

Number of Cyware IOCs By TLP

Displays number of IOCs for each TLP value within the specified IOC Active time filter.

Visualization type: Pie Chart

Number of Cyware IOCs By Severity

Displays number of IOCs for each severity within the specified IOC Active time filter.

Visualization type: Pie Chart

Number of Cyware IOCs By IOC Type

Displays number of IOCs for each IOC Type within the specified IOC Active time filter.

Visualization type: Pie Chart

Timeline of IOCs count by Severity

Displays number of IOCs for each IOC Type within the specified IOC Active time filter.

Visualization type: Line Chart

Number of Cyware IOCs By Country

Displays number of IOCs for each country within the specified IOC Active time filter.

Visualization type: Pie Chart

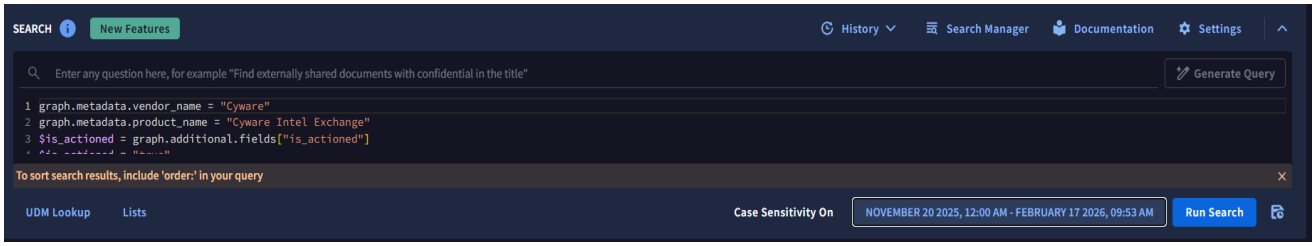
Number of Cyware IOCs By Risk Score

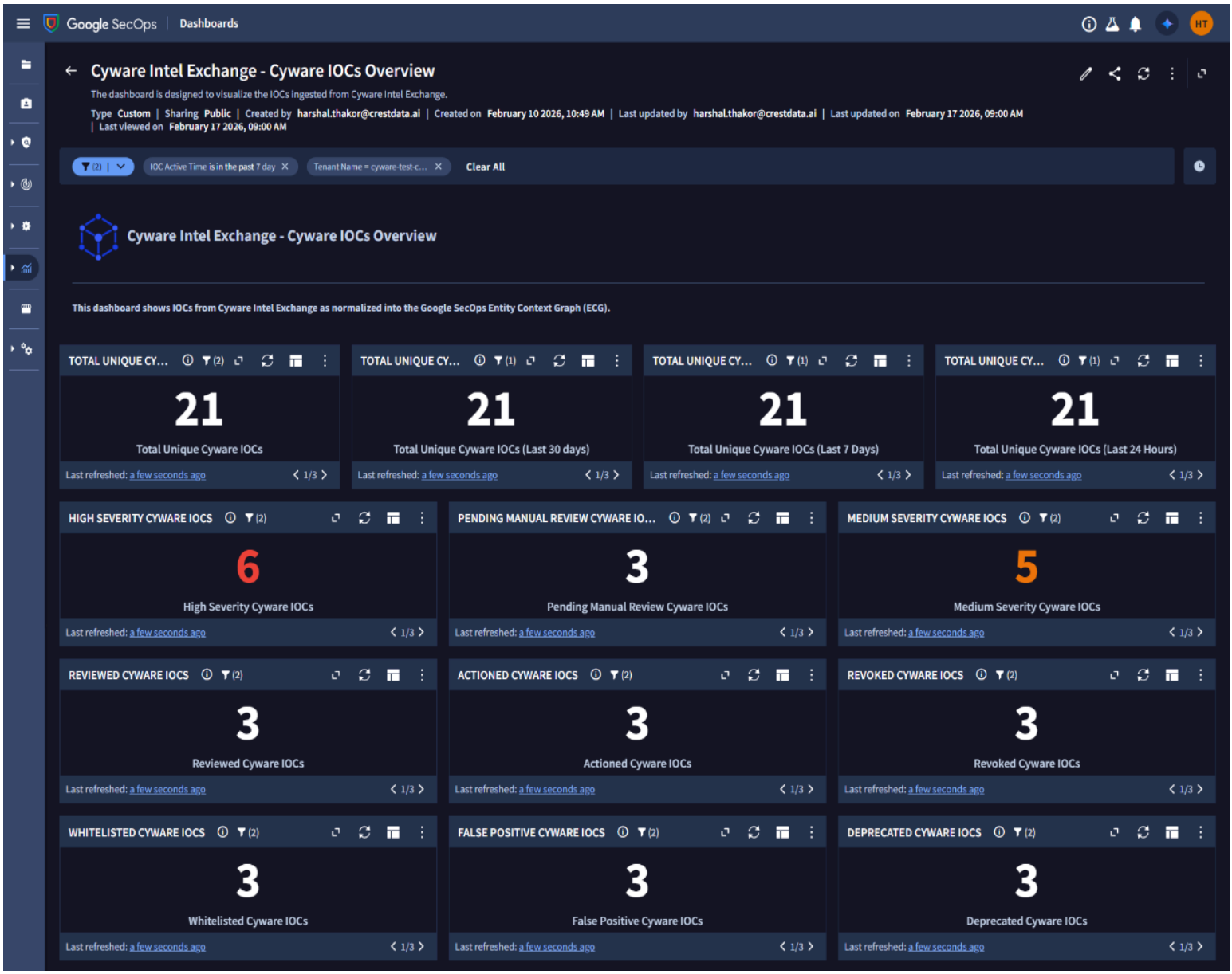
Shows number of IOCs segmented by risk score ranges (None = 0, Low = 1-29, Medium = 30-69, High = 70-100) within the specified IOC Active time filter.

Visualization type: Pie Chart

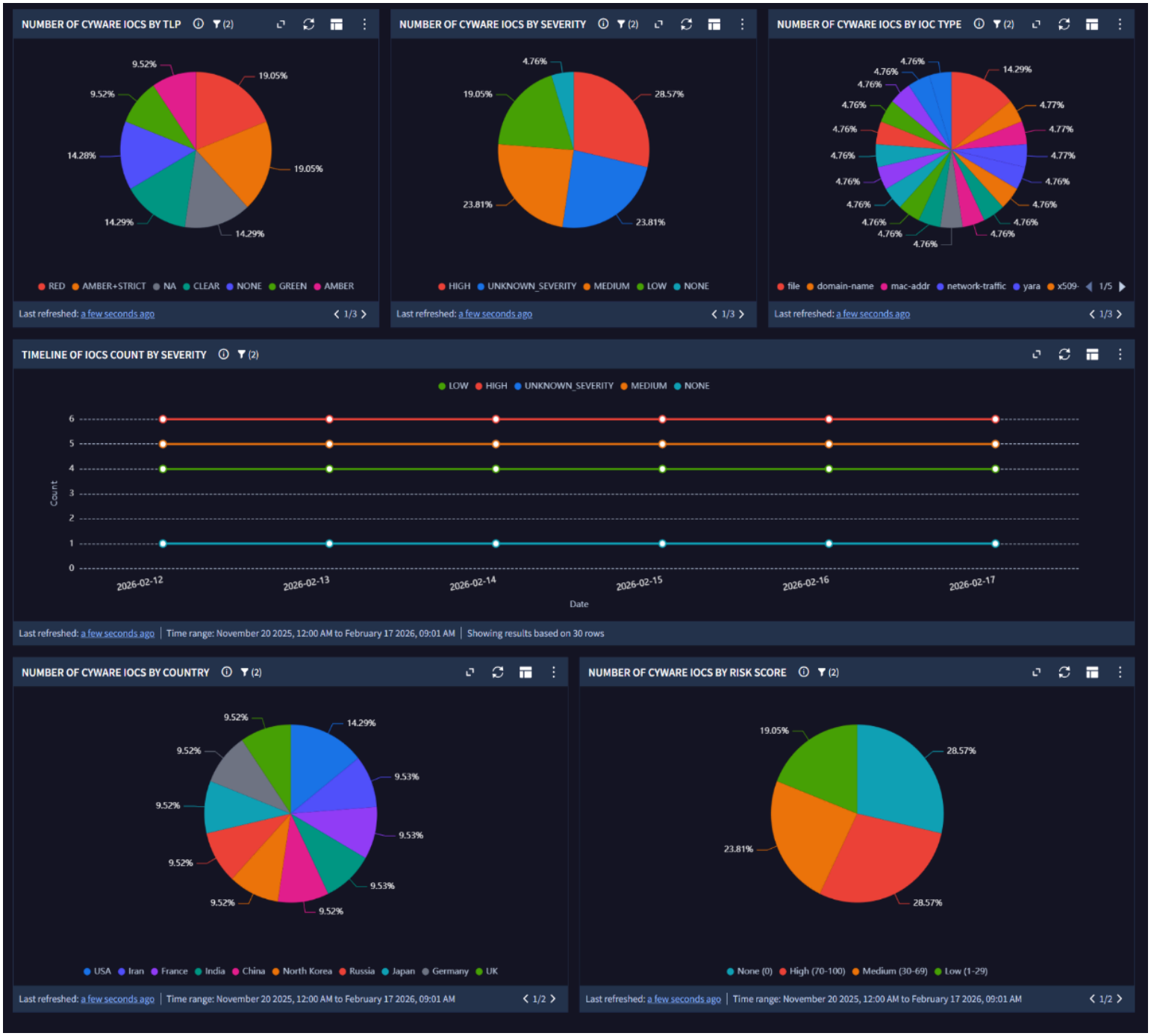
Note:

- For drilldown panels, once the drilldown is triggered, the time range must be manually set to match the time range defined in the IOC Active Time filter within the search. This behavior is due to a limitation in Google SecOps that filter not applied in drilldown query. And keep the Case Sensitivity On.
- If tenant name filter is applied in dashboard then in drilldown, keep the Case Sensitivity On and in first line of drilldown query, add below line with your tenant name:
  `graph.entity.namespace = "your_tenant_name"`





Cyware Intel Exchange - IOCs Overview -1



Cyware Intel Exchange - IOCs Overview -2

Detection Rules

Create Detection Rules

After collecting the data into Google SecOps in the form of entities for the Cyware Indicators, Correlation rules scans Cyware indicators and third-party events ingested into the Google SecOps SIEM to generate detections and alerts.

Create & Enable Detection Rule

1. From Google SecOps SIEM, navigate to **Detection > Rules & Detections**.
2. Go to the **Rules Editor** Tab**,** click on the **NEW** button**.**
3. Copy and paste the Rule in editor.
4. Click on **SAVE NEW RULE.**
5. Click on 3 dots on the top right of the editor.
6. To generate alerts, Enable **Alerting**
7. Enable **Live Rule** to activate Detection Rule**.**

See Google SecOps: Detection Rules for more information.

Detection Rules

1. **Cyware Intel Exchange – Brute Force Attempt**

   Description: This high-severity rule detects multiple login attempts from IP addresses flagged as malicious by Cyware Intel Exchange over a defined time window . It triggers when user login events show repeated login attempts originating from a Cyware Intel Exchange classified IP with high risk score.

   Rule:

```
rule cyware_intel_exchange_brute_force_attempt {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Brute Force Login Detection"
    description = "Detects multiple failed login attempts from IPs
flagged by Cyware threat intelligence."
    severity = "High"
    priority = "Critical"
```

```
    mitre_attack_tactic = "Credential Access"
    mitre_attack_technique = "T1110 - Brute Force"
    tags = "cloud security, threat intelligence, cyware intel exchange"

  events:
    $login.metadata.event_type = "USER_LOGIN"
    $login.security_result.action = "BLOCK" or
    $login.security_result.action = "FAIL"
    (
        strings.to_lower($login.src.ip) = $correlation_ip or
        strings.to_lower($login.principal.ip) = $correlation_ip
    )
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "IP_ADDRESS"
    $cyware.graph.entity.ip = $correlation_ip
  match:
    $correlation_ip over 15m

  outcome:
    $principal_ip = array_distinct($login.principal.ip)
    $principal_hostname = array_distinct($login.principal.hostname)
    $principal_user_userid = array_distinct($login.principal.user.userid)
    $principal_mac = array_distinct($login.principal.mac)

    $event_count = count_distinct($login.metadata.id)

    $source_ip = array_distinct($login.src.ip)
    $source_hostname = array_distinct($login.src.hostname)
    $source_user_userid = array_distinct($login.src.user.userid)
    $source_mac = array_distinct($login.src.mac)

    // Cyware-specific metadata - corrected field paths
    $cyware_threat_id =
array_distinct($cyware.graph.metadata.threat.threat_id)
    $cyware_risk_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $cyware_severity =
array_distinct($cyware.graph.metadata.threat.severity)
    $cyware_description =
array_distinct($cyware.graph.metadata.description)
```

```
    $cyware_tags = array_distinct($cyware.graph.entity.artifact.tags)
    $cyware_country =
array_distinct($cyware.graph.entity.artifact.location.country_or_region)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
        $login and $cyware and ($event_count > 5)
}
```

2. Cyware Intel Exchange – Malicious Domain Detection

Description: This high-priority rule detects DNS queries attempting to resolve domains marked as malicious by Cyware Intel Exchange. It triggers when Network DNS logs show queries for domains classified as malicious on Cyware Intel Exchange with a high risk score.

Rule:

```
rule cyware_intel_exchange_malicious_domain_detection {

  meta:
    author = "Cyware Intel Exchange"
    rule_name = "Cyware Intelligence - Malicious Domain DNS Resolution
Detection"
    description = "Detects DNS queries for domains marked as malicious by
Cyware threat intelligence."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Command and Control"
    mitre_attack_technique = "T1071.004 - Application Layer Protocol:
DNS"
    tags = "cloud security, threat intelligence, cyware intel exchange,
dns"

  events:
    $dns.metadata.event_type = "NETWORK_DNS"
    (
        strings.to_lower($dns.principal.hostname) = $correlation_domain
or
        strings.to_lower($dns.principal.asset.hostname) =
$correlation_domain or
        strings.to_lower($dns.target.hostname) = $correlation_domain or
```

```
        strings.to_lower($dns.target.asset.hostname) =
$correlation_domain
    )

    // Cyware entity match for domain
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "DOMAIN_NAME"
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.domain.name = $correlation_domain

  match:
    $correlation_domain over 1h

  outcome:
    $principal_ip = array_distinct($dns.principal.ip)
    $principal_asset_hostname =
array_distinct($dns.principal.asset.hostname)
    $principal_hostname = array_distinct($dns.principal.hostname)
    $principal_user_userid = array_distinct($dns.principal.user.userid)

    $source_ip = array_distinct($dns.src.ip)
    $source_hostname = array_distinct($dns.src.hostname)

    $target_ip = array_distinct($dns.target.ip)
    $target_hostname = array_distinct($dns.target.hostname)

    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $dns and $cyware
}
```

3. Cyware Intel Exchange – Malicious Email Sender Detection

Description: This rule detects emails in email transaction events that are classified as malicious with high risk score by Cyware Intel Exchange.

Rule:

```
rule cyware_intel_exchange_malicious_email_sender_detection {

  meta:
```

```
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious Email Sender Detection"
    description = "Detects inbound emails from sender addresses or
domains classified as malicious by Cyware."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Initial Access"
    mitre_attack_technique = "T1566 - Phishing"
    tags = "cloud security, threat intelligence, cyware intel exchange,
email"

  events:
    $email.metadata.event_type = "EMAIL_TRANSACTION"
    (
        strings.to_lower($email.network.email.to)  = $correlation_email
or
        strings.to_lower($email.network.email.from)  = $correlation_email
or
        strings.to_lower($email.network.email.mail_id)  =
$correlation_email
    )

    // Cyware entity match for email
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    (
        $cyware.graph.metadata.entity_type = "RESOURCE" or
        $cyware.graph.metadata.entity_type = "USER"
    )
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.user.email_addresses = $correlation_email

  match:
    $correlation_email over 1h

  outcome:
    $email_from = array_distinct($email.network.email.from)
    $email_to = array_distinct($email.network.email.to)
    $email_subject = array_distinct($email.network.email.subject)
    $email_cc = array_distinct($email.network.email.cc)
    $email_bcc = array_distinct($email.network.email.bcc)
```

```
    $principal_ip = array_distinct($email.principal.ip)
    $principal_hostname = array_distinct($email.principal.hostname)

    $source_ip = array_distinct($email.src.ip)
    $source_hostname = array_distinct($email.src.hostname)

    $target_ip = array_distinct($email.target.ip)
    $target_user_email =
array_distinct($email.target.user.email_addresses)

    $cyware_risk_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $email and $cyware
}
```

4. **Cyware Intel Exchange – Malicious File Hash md5 Detection**

   Description: Detects file hashes of type MD5 classified as malicious by Cyware Intel Exchange in network telemetry. It triggers when a file hash in logs matches a file hash in Cyware Intel Exchange with a high risk score .

   Rule:

```
rule cyware_intel_exchange_malicious_file_hash_md5_detection {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious File Hash Detection"
    description = "Detects file hashes classified as malicious by Cyware
in network telemetry."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Execution, Lateral Movement"
    mitre_attack_technique = "T1204 - User Execution, T1570 - Lateral
Tool Transfer"
    tags = "cloud security, threat intelligence, cyware intel exchange,
malware"

  events:
    (
```

```
        $file.metadata.event_type = "FILE_CREATION" or
        $file.metadata.event_type = "FILE_MODIFICATION" or
        $file.metadata.event_type = "FILE_COPY" or
        $file.metadata.event_type = "FILE_OPEN"
    )
    (
        strings.to_lower($file.target.file.md5) = $correlation_hash or
        strings.to_lower($file.src.file.md5) = $correlation_hash or
        strings.to_lower($file.principal.file.md5) = $correlation_hash or
        strings.to_lower($file.principal.process.file.md5) =
$correlation_hash
    )

    // Cyware entity match for file hash
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "FILE"
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.file.md5 = $correlation_hash

  match:
    $correlation_hash over 1h

  outcome:
    $principal_file_md5 =
array_distinct($file.principal.process.file.md5)
    $principal_asset_hostname =
array_distinct($file.principal.asset.hostname)
    $principal_asset_id = array_distinct($file.principal.asset.asset_id)
    $principal_user_userid = array_distinct($file.principal.user.userid)

    $file_path = array_distinct($file.target.file.full_path)
    $file_md5 = array_distinct($file.target.file.md5)

    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $file and $cyware
}
```

5. **Cyware Intel Exchange – Malicious File Hash sha1 Detection**

Description: Detects file hashes of type SHA-1 classified as malicious by Cyware Intel Exchange in network telemetry. It triggers when a file hash in logs matches a file hash in Cyware Intel Exchange with a high risk score.

Rule:

```
rule cyware_intel_exchange_malicious_file_hash_sha1_detection {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious File Hash Detection"
    description = "Detects file hashes classified as malicious by Cyware
in network telemetry."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Execution, Lateral Movement"
    mitre_attack_technique = "T1204 - User Execution, T1570 - Lateral
Tool Transfer"
    tags = "cloud security, threat intelligence, cyware intel exchange,
malware"

  events:
    (
        $file.metadata.event_type = "FILE_CREATION" or
        $file.metadata.event_type = "FILE_MODIFICATION" or
        $file.metadata.event_type = "FILE_COPY" or
        $file.metadata.event_type = "FILE_OPEN"
    )
    (
        strings.to_lower($file.target.file.sha1) = $correlation_hash or
        strings.to_lower($file.src.file.sha1) = $correlation_hash or
        strings.to_lower($file.principal.file.sha1) = $correlation_hash
or
        strings.to_lower($file.principal.process.file.sha1) =
$correlation_hash
    )

    // Cyware entity match for file hash
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "FILE"
    $cyware.graph.metadata.threat.risk_score >= 70
```

```
    $cyware.graph.entity.file.sha1 = $correlation_hash

  match:
    $correlation_hash over 1h

  outcome:
    $principal_file_sha1 =
array_distinct($file.principal.process.file.sha1)
    $principal_asset_hostname =
array_distinct($file.principal.asset.hostname)
    $principal_asset_id = array_distinct($file.principal.asset.asset_id)
    $principal_user_userid = array_distinct($file.principal.user.userid)

    $file_path = array_distinct($file.target.file.full_path)
    $file_sha1 = array_distinct($file.target.file.sha1)

    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $file and $cyware
}
```

6. **Cyware Intel Exchange – Malicious File Hash sha256 Detection**

   Description: Detects file hashes of type SHA-256 classified as malicious by Cyware Intel Exchange in network telemetry. It triggers when a file hash in logs matches a file hash in Cyware Intel Exchange with a high risk score.

   Rule:

```
rule cyware_intel_exchange_malicious_file_hash_sha256_detection {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious File Hash Detection"
    description = "Detects file hashes classified as malicious by Cyware
in network telemetry."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Execution, Lateral Movement"
    mitre_attack_technique = "T1204 - User Execution, T1570 - Lateral
Tool Transfer"
```

```
    tags = "cloud security, threat intelligence, cyware intel exchange,
malware"

  events:
    (
        $file.metadata.event_type = "FILE_CREATION" or
        $file.metadata.event_type = "FILE_MODIFICATION" or
        $file.metadata.event_type = "FILE_COPY" or
        $file.metadata.event_type = "FILE_OPEN"
    )
    (
        strings.to_lower($file.target.file.sha256) = $correlation_hash or
        strings.to_lower($file.src.file.sha256) = $correlation_hash or
        strings.to_lower($file.principal.file.sha256) = $correlation_hash
or
        strings.to_lower($file.principal.process.file.sha256) =
$correlation_hash
    )

    // Cyware entity match for file hash
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "FILE"
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.file.sha256 = $correlation_hash

  match:
    $correlation_hash over 1h

  outcome:
    $principal_file_sha256 =
array_distinct($file.principal.process.file.sha256)
    $principal_asset_hostname =
array_distinct($file.principal.asset.hostname)
    $principal_asset_id = array_distinct($file.principal.asset.asset_id)
    $principal_user_userid = array_distinct($file.principal.user.userid)

    $file_path = array_distinct($file.target.file.full_path)
    $file_sha256 = array_distinct($file.target.file.sha256)

    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
```

```
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $file and $cyware
}
```

7. **Cyware Intel Exchange – Malicious IP Detection**

   Description: Detects network traffic to/from IP addresses identified as malicious by Cyware Intel Exchange.

   Rule:

```
rule cyware_intel_exchange_malicious_ip_detection {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious IP Communication
Detection"
    description = "Detects network traffic to/from IP addresses
identified as malicious by Cyware."
    severity = "High"
    priority = "Critical"
    mitre_attack_tactic = "Command and Control, Initial Access"
    mitre_attack_technique = "T1071 - Application Layer Protocol, T1190 -
Exploit Public-Facing Application"
    tags = "cloud security, threat intelligence, cyware intel exchange"

  events:
    $network.metadata.event_type = "NETWORK_CONNECTION"
    (
        strings.to_lower($network.src.ip) = $correlation_ip or
        strings.to_lower($network.principal.ip) = $correlation_ip or
        strings.to_lower($network.target.ip) = $correlation_ip
    )

    // Cyware entity match for IP
    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "IP_ADDRESS"
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.ip = $correlation_ip

  match:
    $correlation_ip over 1h
```

```
  outcome:
    $principal_ip = array_distinct($network.principal.ip)
    $principal_hostname = array_distinct($network.principal.hostname)
    $principal_user_userid =
array_distinct($network.principal.user.userid)
    $principal_mac = array_distinct($network.principal.mac)

    $source_ip = array_distinct($network.src.ip)
    $source_hostname = array_distinct($network.src.hostname)
    $source_mac = array_distinct($network.src.mac)
    $source_port = array_distinct($network.src.port)

    $target_ip = array_distinct($network.target.ip)
    $target_hostname = array_distinct($network.target.hostname)
    $target_mac = array_distinct($network.target.mac)
    $target_port = array_distinct($network.target.port)

    $network_protocol = array_distinct($network.network.ip_protocol)

    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $network and $cyware
}
```

8. **Cyware Intel Exchange – Malicious URL Access Detection**

   Description: This rule identifies network connections that attempt to access URLs marked as malicious by Cyware Intel Exchange. It triggers when network connection or network http events show a destination URL that matches Cyware Intel Exchange.

   Rule:

```
rule cyware_intel_exchange_malicious_url_access_detection {

  meta:
    author = "CYWARE INTEL EXCHANGE"
    rule_name = "Cyware Intelligence - Malicious URL Access Detection"
    description = "Detects network connections attempting to access URLs
marked as malicious by Cyware."
    severity = "High"
```

```
    priority = "Critical"
    mitre_attack_tactic = "Initial Access, Command and Control"
    mitre_attack_technique = "T1566 - Phishing, T1071 - Application Layer
Protocol"
    tags = "cloud security, threat intelligence, cyware intel exchange"

  events:
    (
        $network.metadata.event_type = "NETWORK_HTTP" or
        $network.metadata.event_type = "NETWORK_CONNECTION"
    )
    (
        strings.to_lower($network.target.url) = $correlation_url or
        strings.to_lower($network.principal.url) = $correlation_url or
        re.regex(strings.to_lower($network.network.http.referral_url),
$correlation_url)
    )

    $cyware.graph.metadata.event_metadata.base_labels.log_types  = "CTIX"
    $cyware.graph.metadata.product_name = "Cyware Intel Exchange"
    $cyware.graph.metadata.entity_type = "URL"
    $cyware.graph.metadata.threat.risk_score >= 70
    $cyware.graph.entity.url = $correlation_url

  match:
    $correlation_url over 1h

  outcome:
    $principal_ip = array_distinct($network.principal.url)
    $principal_hostname = array_distinct($network.principal.hostname)
    $principal_user_userid =
array_distinct($network.principal.user.userid)

    $source_ip = array_distinct($network.src.url)
    $source_hostname = array_distinct($network.src.hostname)

    $target_ip = array_distinct($network.target.ip)
    $target_hostname = array_distinct($network.target.hostname)
    $target_url = array_distinct($network.target.url)

    $http_user_agent = array_distinct($network.network.http.user_agent)
    $http_method = array_distinct($network.network.http.method)
```

```
    $cyware_confidence_score =
array_distinct($cyware.graph.metadata.threat.risk_score)
    $risk_score = max($cyware.graph.metadata.threat.risk_score)

  condition:
    $network and $cyware
}
```

Locate Environment Variables

CHRONICLE_CUSTOMER_ID

Steps to find:

1. Log in to the GCP console (console.cloud.google.com).
2. From the GCP Navigation Menu, Navigate to Security > Google SecOps.
3. Expand Instance details.
4. Copy the Customer ID value.

CHRONICLE_REGION

Steps to find:

1. Log in to the GCP console (console.cloud.google.com).
2. Navigate to Security > Google SecOps.
3. Expand Instance details.
4. Copy the Region value.

CHRONICLE_PROJECT_NUMBER

Steps to find:

1. Log in to the GCP console (console.cloud.google.com).
2. On the home page you can find Project number.

GCP_BUCKET_NAME

Steps to find:

1. Log in to the GCP console (console.cloud.google.com).
2. From the GCP Navigation Menu, Cloud Storage > Buckets.
3. Copy the bucket name (e.g., cyware-bucket).

CYWARE_ACCESS_ID, CYWARE_SECRET_KEY, CYWARE_BASE_URL

Steps to find:

1. Go to the Cyware Intel Exchange instance.
2. Click on Administration icon from navbar.
3. Go to Integration Management → THIRD PARTY DEVELOPERS → CTIX Integrators.
4. Click on Add New.
5. Enter name, description and select expiration.
6. Click on Add New.
7. Copy the Access ID, Secret Key and Endpoint URL.
   (Where Endpoint URL is CYWARE_BASE_URL)

Limitations

1. If the user does not specify the required environment variable while configuring the Cloud Function, the script deployment will fail.
2. CBN parser will only be able to parse the Cyware Intel Exchange entity data.
3. We suggest using the second generation of Cloud Function. The first generation of Cloud Function has a maximum execution time of 9 minutes and the second generation of Cloud Function has a maximum execution time of 60 minutes. If the execution time of the Cloud Function exceeds timeout then there are chances that the complete data won't be ingested in the Google SecOps.
4. The Google SecOps Ingestion API has a payload limit of 4 MB. Logs exceeding this limit will not be ingested and will be skipped. To minimize data loss, please ensure that log sizes remain within the allowed limit.
5. The comma-separated multiple values format for CYWARE_SAVED_RESULT_SET_NAME is not supported during manual Cloud Function deployment. To use comma-separated values, please deploy the function using the provided deployment script.

6. Google SecOps does not enforce validation on time range filters. Users can configure a start time greater than the end time, or set both values to be identical, without receiving an error. Filter validation logic is platform managed and cannot be extended.
7. We do not have any marketplace for Google SecOps integration, so as a part of deliverables, we use to push our integration code to google public repo which undergoes many checks from google side.
8. Time range granularity in Google SecOps follows calendar-based logic. Selecting "Past 1 Week" on a Monday considers the Monday-to-Sunday range, potentially returning only the current day's data if selected at the start of a week. Similar behavior applies to monthly filters.
9. We recommend setting the timeout in the RUNTIME variable to the maximum value (3600) to prevent the Cloud Function from terminating during data collection.
10. The dashboard displays data in a tabular format and will show only 10000 records.
11. In Google SecOps, raw log data is not directly accessible. This creates a limitation when building dashboards, as the visualization layer can only rely on processed or parsed data. As a result, the dashboards cannot always be fully populated with granular event details.
12. In Google SecOps Dashboard, the count displayed in a Single Value panel may not match the entrires shown in Table panels. This is expected behavior, as the Single Value panel displays unique entries, while Table panels display all ingested entries for entities.
13. The activeness of an IOC Entity in Google SecOps is determined by its interval.start_time and interval.end_time fields. This is by design and documented behavior. The system uses the explicitly defined time interval to determine the entity's active period. As a result, users could see the entity in the dashboard for a larger period of time as the IOC entity is active. Also, users will see past ingested entities outside of the selected time range filter because those entities are still active. For more details: Google SecOps Community discussion.
14. The correlation dashboard is limited to support ip address, domain and file hash (MD5, SHA-1, SHA-256) only due to Google SecOps limitation.
15. To get accurate dashboard data, provide a time range filter greater than the past 13 hours.
16. For the parser, The enrichment_data[].tool_response field is stored as a string in additional.fields, while its top-level key-value pairs are added as individual fields with the naming pattern

additional.fields["enrichment_data_%{index}_tool_response_%{key}"]. Nested objects within tool_response are not flattened, only included in the full string.

17. For the parser, file-type indicators with subtypes other than MD5, SHA-1, and SHA-256 are mapped to the resource schema, as the UDM's file schema does not provide corresponding fields for those subtypes. Refer link

Known Behaviors

1. In Google SecOps, when applying time filters on the UDM Dashboard, there is a mismatch in results between the Dashboard time filter and the UDM Search time filter. Both filters are configured on the same field (graph.metadata.interval.start_time) and use the exact same time range. However, the counts returned are inconsistent.
2. Newly ingested entities often take additional time to appear on the dashboard. This latency impacts real-time monitoring and reduces the effectiveness of dashboards for time-sensitive investigations. A support ticket was raised regarding this issue for further investigation and resolution.
3. When filters are applied in Google SecOps dashboards, the drilldown functionality does not work as expected. Specifically, the timeframe displayed in drilldown results

differs from the timeframe shown in the filtered results, leading to inconsistencies and misalignment between the two views.

4. When Cloud Run functions execute for more than 30 minutes, Cloud Scheduler will show a "Failed" status with 504 Gateway Timeout errors. This is expected behavior and does not indicate actual function failure. The Cloud Run function continues execution despite the timeout in Cloud Scheduler.
   Error Message Example:
   ```
   ERROR <timestamp> [httpRequest.requestMethod:POST]
   [httpRequest.status: 504] [httpRequest.responseSize: 72 B]
   [httpRequest.latency:1,799.798 s]
   [httpRequest.userAgent:Google-Cloud-Scheduler]
   https://<cloud_function_uri>
   ```

Troubleshooting

This section describes the common issues that might happen during the deployment or the running of the app and the steps to resolve the issues.

1. GCloud logs can be used for troubleshooting.
    a. Log in to the "https://console.cloud.google.com/" using valid credentials.
    b. Navigate to 'Cloud functions' and click on the deployed function where you can find the logs module.
    c. Logs can be filtered using severity.
2. If you test the cloud function immediately after deploying it on gcloud, It might be possible that the cloud function will not work as expected. To resolve this, wait for a few seconds and then test it.
3. If the cloud function stops its execution because memory exceeds the limit, reconfigure the cloud function's memory configuration and increase the memory limit.
4. The dashboard may be slow to load or unresponsive - This could be due to a problem with the data source being unavailable or having too much data, the query that is being used, or the way that the dashboard is being rendered.
5. While executing deployment script with ./cyware_deploy.sh, if error mentioned below occurred then run the command: `sed -i 's/\r//g' cyware_deploy.sh` to fix the error then execute the script.

```
harshal_thakor@cloudshell:~ ([redacted])$ ./cyware_deploy.sh
./cyware_deploy.sh: line 13: $'\r': command not found
./cyware_deploy.sh: line 29: $'\r': command not found
./cyware_deploy.sh: line 30: syntax error near unexpected token `$'{\r''
'/cyware_deploy.sh: line 30: `print_message() {
```

6. Entities are successfully ingested but not displayed in search results this could be due to following reasons.
    a. The time range for the search is outside of the timestamp associated with the ingested event.
    b. The ingested entity might be a duplicate entity with the same payload.

GCP Resources/Services Approximate Cost Details

Service	Standard Configurations	Purpose	Reference
Cloud Functions	Type: Memory: 8192MB CPU: 4.8GHz Execution time per function (ms): 3600 Invocations per month: 1500 Minimum number of instances: 1	Function / Script which pulls data from Cyware Intel Exchange Platform using API and ingest into Google SecOps.	Approx cost ~ $66/month https://cloud.google.com/functions/pricing
Cloud Storage (Bucket)	Total Amount of Storage: 1 GiB	Storage bucket used to manage API checkpoints	Approx cost ~ $0.02/month https://cloud.google.com/storage/pricing
Secret Manager	Access operations: 1500	Used to maintain credentials.	Approx cost ~ $0/month https://cloud.google.com/secret-manager/pricing
Cloud Scheduler	Total amount of jobs: 1	Scheduler which executes above cloud function at a specific time interval.	Approx cost ~ $0/month https://cloud.google.com/scheduler/pricing

Note: Users can also calculate (using pricing calculator) the estimated price of the Google Cloud services used.

References

- Cloud Function
- Cloud Scheduler
- Cloud Secret Manager
- Install the gcloud CLI
- Deploying cloud functions from local machine
- Google SecOps Ingestion API Payload Limit
- Cyware Intel Exchange API Documentation